Dark matter and the neutrino background with the full lattice response of cryogenic crystals

Derivations, implementation and results of the NuFog v2 pipeline

Israel Hernández

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Abstract

This report documents the calculation of dark matter (DM) discovery limits in the presence of the coherent elastic neutrino–nucleus scattering (CEvNS) background for crystal targets with energy thresholds from tens of meV to eV. In that range the recoil energy is absorbed by the lattice through the emission of one or many phonons, not by a free nucleus. Both the DM signal and the neutrino background are therefore governed by the dynamic structure factor $S(\vec{q},\omega)$ of the crystal.

All formulas are derived from first principles:

- the scattering rate from Fermi's golden rule;
- the multiphonon expansion of a harmonic crystal in the incoherent approximation, with its sum rules and impulse limit;
- the DM and CEvNS rates, the Asimov discovery test with flux nuisance parameters, and the scaling laws that distinguish a background from a "fog";
- the directional (daily-modulation) signal of anisotropic crystals.

The implementation and its validation are described. Results are given for $CaWO_4$, Al_2O_3, TeO_2, $LiNbO_3$ and Si, together with a comparison with CRESST and with the literature. A final section derives why the lattice response must be known precisely: the low-mass rate depends on the phonon spectrum as a high power of its moments, and the ability to separate DM from neutrinos rests on the spectral shape that $S(\vec{q},\omega)$ produces.

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1 Scope and conventions

We use natural units $\hbar = c = k_B = 1$. The DM mass is m_χ, the nucleon mass m_n, and the DM–nucleon reduced mass $\mu = m_\chi m_n/(m_\chi + m_n)$. Atoms in the unit cell are labelled by d, with mass m_d, mass number A_d, atomic number Z_d and displacement $\vec{u}_d$. The momentum transfer is $\vec{q} = q\,\hat{q}$, and the energy deposited in the crystal is ω. V is the unit-cell volume, $M_{\rm cell} = \sum_d m_d$ the cell mass, and $\rho_T = M_{\rm cell}/V$ the target mass density.

The analysis assumes:

- spin-independent DM–nucleon scattering through a heavy mediator, with equal couplings to protons and neutrons;
- the Standard Halo Model (SHM) with $\rho_\chi = 0.3\,\mathrm{GeV\,cm^{-3}}$, $v_0 = 220$, $v_E = 232$ and $v_{\rm esc} = 544\,\mathrm{km\,s^{-1}}$;
- a crystal at zero temperature.

The detector response is a threshold $E_{\rm th}$ and a Gaussian resolution $\sigma_E = E_{\rm th}/5$, following the CRESST convention.

2 Scattering off a crystal: the dynamic structure factor

2.1 Rate from Fermi's golden rule

For a contact interaction, the DM scatters off each nucleus J with an amplitude proportional to its coherent coupling f_J ($f_J = A_J$ for spin-independent scattering) times the nuclear form factor $F_J(q)$. The normalization is chosen such that a single free nucleon has cross section σ_n. In Born approximation the transition rate of a DM particle with velocity $\vec{v}$, which excites the crystal from its ground state $|0\rangle$ to a state $|f\rangle$, is

$$\Gamma(\vec{v}) = \frac{\pi\sigma_n}{\mu^2}\int \frac{\mathrm{d}^3q}{(2\pi)^3}\,S(\vec{q},\omega_{\vec{q}}), \qquad \omega_{\vec{q}} = \vec{q}\cdot\vec{v} - \frac{q^2}{2m_\chi}, \tag{2.1}$$

where energy conservation fixes the deposited energy $\omega_{\vec{q}}$, and the *dynamic structure factor* is

$$S(\vec{q},\omega) = \frac{2\pi}{V_{\rm tot}}\sum_f \left|\langle f|\sum_J f_J F_J(q)\,e^{i\vec{q}\cdot\vec{r}_J}|0\rangle\right|^2 \delta(E_f - E_0 - \omega). \tag{2.2}$$

Here $V_{\rm tot}$ is the crystal volume. Because S is extensive in the number of cells, it is convenient to quote it per unit cell, replacing $V_{\rm tot}$ by V and the sum over J by a sum over the atoms d of one cell (Sec. 3.2).

2.2 Differential rate per unit target mass

The event rate per unit target mass is

$$R = \frac{1}{\rho_T}\frac{\rho_\chi}{m_\chi}\int \mathrm{d}^3v\, f(\vec{v})\,\Gamma(\vec{v}). \tag{2.3}$$

Inserting $1 = \int \mathrm{d}\omega\,\delta(\omega - \omega_{\vec{q}})$,

$$\frac{\mathrm{d}R}{\mathrm{d}\omega} = \frac{1}{\rho_T}\frac{\rho_\chi}{m_\chi}\frac{\pi\sigma_n}{\mu^2}\int\frac{\mathrm{d}^3q}{(2\pi)^3}\,S(\vec{q},\omega)\int \mathrm{d}^3v\, f(\vec{v})\,\delta\!\left(\vec{q}\cdot\vec{v} - \frac{q^2}{2m_\chi} - \omega\right). \tag{2.4}$$

The velocity integral defines the *Radon transform* of f,

$$\int \mathrm{d}^3v\, f(\vec{v})\,\delta(\vec{q}\cdot\vec{v} - qv^*) = \frac{1}{q}\,g(v^*,\hat{q}), \qquad g(w,\hat{q}) \equiv \int \mathrm{d}^3v\, f(\vec{v})\,\delta(\vec{v}\cdot\hat{q} - w), \qquad v^* = \frac{q}{2m_\chi} + \frac{\omega}{q}. \tag{2.5}$$

Here v^* is the minimum velocity component along $\hat{q}$ that can deposit ω with momentum q.

Isotropic response. If S depends only on $q = |\vec{q}|$, the angular integral of g can be done first. Using $\mathrm{d}^3q = q^2 \mathrm{d}q\, \mathrm{d}\Omega_q$ and

$$\int \frac{\mathrm{d}\Omega_q}{4\pi}\, g(w, \hat{q}) = \int \mathrm{d}^3v\, f(\vec{v}) \int \frac{\mathrm{d}\Omega_q}{4\pi}\, \delta(v\cos\theta - w) = \int \mathrm{d}^3v\, \frac{f(\vec{v})}{2v}\, \Theta(v - w) \equiv \frac{1}{2}\, \eta(w), \tag{2.6}$$

where $\eta(v_{\min}) = \int \mathrm{d}^3v\, f(\vec{v})\, \Theta(v - v_{\min})/v$ is the mean inverse speed, one obtains

$$\boxed{\frac{\mathrm{d}R}{\mathrm{d}\omega} = \frac{1}{\rho_T} \frac{\rho_\chi}{m_\chi} \frac{\sigma_n}{4\pi\mu^2} \int q\, \mathrm{d}q\, \eta(v^*)\, S(q, \omega)} \tag{2.7}$$

The steps are $\frac{\pi}{(2\pi)^3} \cdot 4\pi \cdot \frac{1}{2} = \frac{1}{4\pi}$, with one power of q cancelled by Eq. (2.5). The q integral runs up to $q_{\max} = 2m_\chi(v_{\rm esc} + v_E)$.

2.3 Free-nucleus limit (consistency check)

For a free nucleus of mass M and coupling A, Eq. (2.2) per nucleus gives $S = (2\pi/V)A^2F^2\, \delta(\omega - q^2/2M)$. Using $\int q\, \mathrm{d}q\, \delta(\omega - q^2/2M) = M$ and $1/(\rho_T V) = 1/M_{\rm cell}$ (one nucleus per "cell"),

$$\frac{\mathrm{d}R}{\mathrm{d}E_R} = \frac{\rho_\chi}{m_\chi} \frac{\sigma_n A^2 F^2(q)}{2\mu^2}\, \eta(v_{\min}), \qquad v_{\min} = \sqrt{\frac{ME_R}{2\mu_{\chi N}^2}}. \tag{2.8}$$

This is the textbook rate per nucleus. The pipeline reproduces it numerically to 0.1% for all five materials (Sec. 9).

3 The harmonic crystal

3.1 Phonons and the density-of-states tensor

Write the position of atom d in cell ℓ as $\vec{r}_{\ell d} = \vec{R}_\ell + \vec{x}_d + \vec{u}_{\ell d}$. In the harmonic approximation,

$$\vec{u}_{\ell d} = \sum_{\nu,\vec{k}} \frac{1}{\sqrt{2N m_d\, \omega_{\nu\vec{k}}}} \left(\vec{e}_{d,\nu\vec{k}}\, a_{\nu\vec{k}}\, e^{i\vec{k}\cdot\vec{R}_\ell} + \text{h.c.} \right), \tag{3.1}$$

where $\omega_{\nu\vec{k}}$ and $\vec{e}_{d,\nu\vec{k}}$ are the frequencies and eigenvector components of the $3n$ branches, N is the number of cells, and the eigenvectors satisfy the completeness relation $\sum_\nu e^{i*}_{d,\nu\vec{k}} e^{j}_{d,\nu\vec{k}} = \delta^{ij}$ for each atom d. We define the *per-atom density-of-states (DOS) tensor*

$$D_d^{ij}(\omega) = \frac{1}{N} \sum_{\nu,\vec{k}} e^{i*}_{d,\nu\vec{k}}\, e^{j}_{d,\nu\vec{k}}\, \delta(\omega - \omega_{\nu\vec{k}}), \qquad \int \mathrm{d}\omega\, D_d^{ij} = \delta^{ij}, \tag{3.2}$$

and its projection along $\hat{q}$,

$$D_d^{\hat{q}}(\omega) = \hat{q}_i\, D_d^{ij}(\omega)\, \hat{q}_j, \qquad \int \mathrm{d}\omega\, D_d^{\hat{q}} = 1. \tag{3.3}$$

The scalar (isotropic) DOS is $\frac{1}{3}\mathrm{Tr}\, D_d$. The tensor is computed from first-principles force constants (phonopy) on a dense $\vec{k}$ mesh without symmetry reduction, with the non-analytic LO–TO correction when Born charges are available.

3.2 Incoherent approximation

Expanding $|\sum_J \dots|^2$ in Eq. (2.2) produces terms with $J = J'$ and interference terms $J \neq J'$. For $q \gg 2\pi/a$, with a the interatomic distance (a few keV), the interference terms average out. Keeping only $J = J'$ gives

$$S(\vec{q},\omega) = \frac{2\pi}{V} \sum_{d\in\text{cell}} f_d^2 F_d^2(q)\; S_d(\vec{q},\omega), \qquad S_d(\vec{q},\omega) = \sum_f \left|\langle f|e^{i\vec{q}\cdot\vec{u}_d}|0\rangle\right|^2 \delta(E_f - E_0 - \omega). \tag{3.4}$$

This approximation fails at the lowest momentum transfers, where single-phonon scattering is coherent and depends on the phonon dispersion. That affects masses below about 1 MeV and the directional signal (Sec. 8).

3.3 Correlation function and Debye–Waller factor

Writing the energy δ-function as a time integral and using the Heisenberg picture,

$$S_d(\vec{q},\omega) = \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{2\pi}\, e^{i\omega t} \left\langle e^{-i\vec{q}\cdot\vec{u}_d(0)}\, e^{i\vec{q}\cdot\vec{u}_d(t)} \right\rangle. \tag{3.5}$$

For a harmonic system, $X \equiv \vec{q}\cdot\vec{u}_d$ is linear in Gaussian (bosonic) operators, so Bloch's identity $\langle e^{-iX(0)} e^{iX(t)}\rangle = e^{-\langle X^2\rangle}\, e^{\langle X(0)X(t)\rangle}$ holds. At $T = 0$,

$$\langle X(0)X(t)\rangle = \sum_{\nu\vec{k}} \frac{|\vec{q}\cdot\vec{e}_{d,\nu\vec{k}}|^2}{2N m_d\, \omega_{\nu\vec{k}}}\, e^{-i\omega_{\nu\vec{k}} t} = \frac{q^2}{2m_d} \int \mathrm{d}\omega'\, \frac{D_d^{\hat{q}}(\omega')}{\omega'}\, e^{-i\omega' t}. \tag{3.6}$$

At $t = 0$ this defines the Debye–Waller exponent

$$2W_d(\vec{q}) \equiv \langle X^2\rangle = \frac{q^2}{2m_d}\, \langle\frac{1}{\omega}\rangle_d^{\hat{q}}, \qquad \langle\frac{1}{\omega}\rangle_d^{\hat{q}} \equiv \int \mathrm{d}\omega\, \frac{D_d^{\hat{q}}(\omega)}{\omega}, \qquad W_d = \frac{q^2}{4m_d} \langle\frac{1}{\omega}\rangle_d^{\hat{q}}. \tag{3.7}$$

3.4 Multiphonon expansion

Expanding $e^{\langle X(0)X(t)\rangle}$ in powers and Fourier transforming term by term,

$$\boxed{S_d(\vec{q},\omega) = e^{-2W_d(\vec{q})} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{q^2}{2m_d}\right)^n T_{d,n}(\omega), \qquad T_{d,1}(\omega) = \frac{D_d^{\hat{q}}(\omega)}{\omega}, \quad T_{d,n} = T_{d,1} * T_{d,n-1}} \tag{3.8}$$

with $T_{d,0} = \delta(\omega)$ (the elastic term). The convolution follows from

$$\int \frac{\mathrm{d}t}{2\pi} e^{i\omega t} \Big[\int \mathrm{d}\omega'\, T_1(\omega') e^{-i\omega' t} \Big]^n = \int \prod_{k=1}^{n} \mathrm{d}\omega_k\, T_1(\omega_k)\; \delta\Big(\omega - \sum_k \omega_k\Big) = T_n(\omega). \tag{3.9}$$

So the n-th term is the emission of exactly n phonons. In the code the series is summed for $n \geq 1$ (the elastic term deposits no energy), up to $n_{\max} = 60$. The T_n are computed by FFT convolution on a uniform grid up to 3 eV.

3.5 Poisson form and sum rules

Define the normalized single-phonon distribution and the mean phonon number

$$\hat{T}_1(\omega) = \frac{T_1(\omega)}{\langle 1/\omega\rangle}, \quad \int \hat{T}_1 = 1; \qquad \lambda_d \equiv 2W_d = \frac{q^2}{2m_d} \langle\frac{1}{\omega}\rangle. \tag{3.10}$$

Since $T_n = \langle 1/\omega\rangle^n \hat{T}_n$ with $\hat{T}_n$ the n-fold convolution of $\hat{T}_1$, Eq. (3.8) becomes

$$\boxed{S_d(\vec{q},\omega) = \sum_{n\geq 0} \underbrace{e^{-\lambda_d}\frac{\lambda_d^n}{n!}}_{\mathrm{Poisson}(n;\lambda_d)} \hat{T}_{d,n}(\omega)} \tag{3.11}$$

The number of emitted phonons is Poisson-distributed with mean λ_d, and each phonon carries an energy drawn from $\hat{T}_1$. Two exact sum rules follow at once:

$$\int \mathrm{d}\omega\, S_d = \sum_n \mathrm{Pois}(n;\lambda) = 1, \qquad \Rightarrow \int \mathrm{d}\omega \sum_{n\geq 1}(\ldots) = 1 - e^{-2W_d}, \tag{3.12}$$

$$\int \mathrm{d}\omega\, \omega\, S_d = \langle n\rangle\, \langle\omega\rangle_{\hat{T}_1} = \lambda_d \cdot \frac{1}{\langle 1/\omega\rangle} = \frac{q^2}{2m_d}, \qquad \text{(f-sum rule)}. \tag{3.13}$$

Here $\langle\omega\rangle_{\hat{T}_1} = \int \omega\, T_1/\langle 1/\omega\rangle = 1/\langle 1/\omega\rangle$ because $\int D = 1$. The f-sum rule states that, on average, the lattice absorbs exactly the free-recoil energy $q^2/2m_d$. These identities hold only if W_d and $T_{d,1}$ are built from the same D_d. Their violation was the main defect of the v1 code (Sec. 9).

3.6 Impulse approximation

For large q the phonon number λ_d is large and the sum is dominated by $n \approx \lambda_d$. Expanding Eq. (3.6) at short times, $e^{-i\omega' t} \simeq 1 - i\omega' t - \frac{1}{2}\omega'^2 t^2$, gives

$$\langle X(0)X(t)\rangle - \langle X^2\rangle \simeq -i\frac{q^2}{2m_d}t - \frac{q^2\langle\omega\rangle_d}{4m_d}t^2, \qquad \langle\omega\rangle_d = \int \omega D_d^{\hat{q}}\,\mathrm{d}\omega. \tag{3.14}$$

Its Fourier transform is a Gaussian:

$$S_d(q,\omega) \simeq \frac{1}{\sqrt{2\pi\Delta_d^2}} \exp\left[-\frac{(\omega - q^2/2m_d)^2}{2\Delta_d^2}\right], \qquad \Delta_d^2 = \frac{q^2\langle\omega\rangle_d}{2m_d}. \tag{3.15}$$

It is centred on the free-nucleus recoil energy and broadened by the zero-point motion. As $q \to \infty$ it tends to $\delta(\omega - q^2/2m_d)$, the free-nucleus result. The code switches from Eq. (3.8) to the Gaussian at $\lambda_d = 16$, i.e. $q_{\mathrm{thr}} = 4\sqrt{2m_d\,\bar{\omega}_{0,d}}$ with $\bar{\omega}_{0,d} = 1/\langle 1/\omega\rangle_d$.

3.7 Infrared regularization

The tensor is built with Gaussian smearing ($\sigma = 0.5\,\mathrm{meV}$), so the numerical $D(\omega)$ is finite at $\omega \to 0$. Then D/ω diverges logarithmically, and $\langle 1/\omega\rangle$ depends on the grid. A real crystal has acoustic modes with $D \propto \omega^2$ (Debye). Below $\omega_c = 2\,\mathrm{meV}$ the DOS is therefore replaced by $D(\omega_c)(\omega/\omega_c)^2$, continuously matched and renormalized to unit integral. $\langle 1/\omega\rangle$, $\langle\omega\rangle$ and every T_n are then computed from the same regularized T_1, so Eqs. (3.12)–(3.13) hold by construction (to about 1% numerically).

3.8 Couplings, form factor and cell normalization

For SI DM, $f_d = A_d$. For CEvNS, $f_d = Q_{W,d} = N_d - (1 - 4\sin^2\theta_W)Z_d$, with $\sin^2\theta_W = 0.23857$. The Helm form factor is

$$F(q) = \frac{3j_1(qR_n)}{qR_n}e^{-q^2s^2/2}, \quad R_n^2 = c^2 + \tfrac{7}{3}\pi^2a^2 - 5s^2, \quad c = 1.23A^{1/3} - 0.60\ \mathrm{fm},\ a = 0.52,\ s = 0.9\ \mathrm{fm}. \tag{3.16}$$

The rates per unit mass contain $(2\pi/V)/\rho_T = 2\pi/M_{\mathrm{cell}}$. The density must therefore be the one implied by the cell used for S, $\rho_T = M_{\mathrm{cell}}/V$, so that V cancels exactly. A tabulated density inconsistent with V biases the rate (by up to 9% in v1).

4 Dark-matter rate

Combining Eqs. (2.7), (3.4) and (3.8):

$$\frac{\mathrm{d}R_\chi}{\mathrm{d}\omega} = \frac{2\pi}{M_{\rm cell}}\frac{\rho_\chi}{m_\chi}\frac{\sigma_n}{4\pi\mu^2}\int_0^{q_{\max}} q\,\mathrm{d}q\;\eta\!\left(\frac{q}{2m_\chi}+\frac{\omega}{q}\right)\sum_d A_d^2 F_d^2(q)\,S_d(q,\omega). \tag{4.1}$$

For the SHM, a Maxwellian truncated at $v_{\rm esc}$ and boosted by v_E, the mean inverse speed is

$$\eta(x) = \frac{1}{2N_{\rm esc}v_E}\times\begin{cases}\mathrm{erf}\frac{x+v_E}{v_0}-\mathrm{erf}\frac{x-v_E}{v_0}-\frac{4v_E}{\sqrt{\pi}v_0}e^{-z^2}, & x<v_{\rm esc}-v_E,\\ \mathrm{erf}\,z-\mathrm{erf}\frac{x-v_E}{v_0}-\frac{2(v_{\rm esc}+v_E-x)}{\sqrt{\pi}v_0}e^{-z^2}, & v_{\rm esc}-v_E\le x<v_{\rm esc}+v_E,\\ 0, & \text{otherwise,}\end{cases} \tag{4.2}$$

with $z = v_{\rm esc}/v_0$ and $N_{\rm esc} = \mathrm{erf}z - \frac{2z}{\sqrt{\pi}}e^{-z^2}$. Kinematics restrict the deposited energy to $\omega \le qv_{\max} - q^2/2m_\chi \le \frac{1}{2}m_\chi v_{\max}^2$. For $m_\chi = 1\,\mathrm{MeV}$ this is about 3 eV, so light DM can only be seen through phonons.

5 CEvNS rate

For a free nucleus, $\frac{\mathrm{d}\sigma}{\mathrm{d}E_R} = \frac{G_F^2 M}{4\pi}Q_W^2\big(1-\frac{ME_R}{2E_\nu^2}\big)F^2$. With $q^2 = 2ME_R$, $\frac{\mathrm{d}\sigma}{\mathrm{d}q^2} = \frac{G_F^2}{8\pi}Q_W^2\big(1-\frac{q^2}{4E_\nu^2}\big)F^2$. In a crystal the free-recoil δ-function is replaced by the lattice response:

$$\frac{\mathrm{d}\sigma_d}{\mathrm{d}q^2\,\mathrm{d}\omega} = \frac{G_F^2}{8\pi}Q_{W,d}^2F_d^2\left(1-\frac{q^2}{4E_\nu^2}\right)S_d(q,\omega), \tag{5.1}$$

and the rate per unit mass is

$$\boxed{\frac{\mathrm{d}R_\nu}{\mathrm{d}\omega} = \frac{1}{\rho_T}\frac{G_F^2}{8\pi^2}\int \mathrm{d}E_\nu\frac{\mathrm{d}\Phi}{\mathrm{d}E_\nu}\int_0^{2E_\nu} q\,\mathrm{d}q\left(1-\frac{q^2}{4E_\nu^2}\right)\frac{2\pi}{V}\sum_d Q_{W,d}^2F_d^2S_d(q,\omega)} \tag{5.2}$$

with $\omega \le q$ for a massless neutrino. In the free limit this reduces to the textbook formula. This fixes the convention: the prefactor $G_F^2/8\pi^2$ goes with $Q_W \simeq N$. A prefactor $G_F^2/8\pi^2$ combined with $Q_W \simeq -N/2$ underestimates the rate by a factor 4, and $G_F^2/4\pi$ with $Q_W/2$ (the v1 code) overestimates it by $\pi/2$.

Table 1: Neutrino sources, flux normalizations and relative uncertainties (Gaussian nuisance parameters).

Source	Φ [cm^{-2} s^{-1}]	r_k
^{8}B	5.46×10^6	0.12
hep	7.98×10^3	0.30
Atmospheric	10.5	0.20
DSNB	86	0.50
^{15}O	2.05×10^8	0.17
^{13}N	2.78×10^8	0.15
^{17}F	5.29×10^6	0.20
pp	5.98×10^{10}	0.006
^{7}Be (384 keV)	4.84×10^8	0.06
^{7}Be (861 keV)	4.35×10^9	0.06
pep	1.44×10^8	0.01

6 Detector response

Spectra are integrated into 200 logarithmic bins between 1 meV and 100 keV. A Gaussian resolution σ_E is applied as a bin-migration matrix

$$\mathcal{R}_{ij} = \Big\langle \tfrac{1}{2}\Big[\operatorname{erf}\tfrac{E_j^{\rm hi}-E}{\sqrt{2}\sigma_E} - \operatorname{erf}\tfrac{E_j^{\rm lo}-E}{\sqrt{2}\sigma_E}\Big]\Big\rangle_{E\in\text{bin } i}, \qquad N_j^{\rm rec} = \sum_i N_i^{\rm true}\mathcal{R}_{ij}, \tag{6.1}$$

to the full DM and neutrino spectra before the threshold cut. Only reconstructed bins with lower edge $\geq E_{\rm th}$ are kept. Applying the cut after smearing is essential: upward fluctuations of sub-threshold events then populate the first bins, as they do in a real detector.

7 Statistical method

7.1 Asimov discovery test

Let B_{kb} be the expected events of neutrino source k in bin b (exposure included), s_b the DM events per unit cross section, and θ_k the flux normalizations with Gaussian priors of relative width r_k. The background-only negative log-likelihood is

$$-\ln L_0(\boldsymbol{\theta}) = \sum_b \big[n_b(\boldsymbol{\theta}) - N_b \ln n_b(\boldsymbol{\theta})\big] + \sum_k \frac{(\theta_k-1)^2}{2r_k^2}, \qquad n_b = \sum_k \theta_k B_{kb}. \tag{7.1}$$

The Asimov data set is $N_b = \sum_k B_{kb} + \sigma s_b$. The discovery statistic is

$$q_0(\sigma) = 2\Big[\min_{\boldsymbol{\theta}}\big(-\ln L_0\big) - \big(-\ln L_1\big)\Big], \qquad -\ln L_1 = \sum_b \big[N_b - N_b\ln N_b\big], \tag{7.2}$$

and the 3σ discovery cross section solves $q_0(\sigma) = 9$ (significance $Z = \sqrt{q_0}$).

Convexity. The Hessian of $-\ln L_0$ is $H_{kk'} = \sum_b B_{kb}B_{k'b}N_b/n_b^2 + \delta_{kk'}/r_k^2$, which is positive definite. The profile is therefore a convex problem with a unique minimum. It is solved exactly by a projected Newton iteration ($\theta_k \geq 0$) with backtracking line search, and q_0 is monotonic in σ, so $\sigma_{\rm disc}$ is found by bisection in $\log\sigma$. Bins with zero background use $n_b = 10^{-30}$, as in v1.

7.2 Background or fog? Scaling with exposure

Let X be the exposure, with $B_{kb} = X\beta_{kb}$ and $s_b = X\varepsilon_b$. For large X the Asimov statistic can be linearized around the minimum:

$$q_0 \simeq \sigma^2 X^2\, \boldsymbol{\varepsilon}^{\mathsf T}\,\mathbf{C}^{-1}\boldsymbol{\varepsilon}, \qquad \mathbf{C} = X\,\mathrm{diag}(\boldsymbol{b}) + X^2\,\boldsymbol{\beta}^{\mathsf T}\mathbf{r}^2\boldsymbol{\beta}, \tag{7.3}$$

with $b_b = \sum_k \beta_{kb}$ and $\mathbf{r}^2 = \mathrm{diag}(r_k^2)$. Decompose $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_\parallel + \boldsymbol{\varepsilon}_\perp$, where $\boldsymbol{\varepsilon}_\parallel$ lies in the span of the background templates $\{\boldsymbol{\beta}_k\}$ and $\boldsymbol{\varepsilon}_\perp$ is orthogonal to it in the metric $\mathrm{diag}(\boldsymbol{b})^{-1}$. By the Woodbury identity, as $X\to\infty$,

$$q_0 \;\longrightarrow\; \sigma^2 X\,\boldsymbol{\varepsilon}_\perp^{\mathsf T}\mathrm{diag}(\boldsymbol{b})^{-1}\boldsymbol{\varepsilon}_\perp \;+\; \sigma^2\,\boldsymbol{\varepsilon}_\parallel^{\mathsf T}\big(\boldsymbol{\beta}^{\mathsf T}\mathbf{r}^2\boldsymbol{\beta}\big)^{+}\boldsymbol{\varepsilon}_\parallel. \tag{7.4}$$

This gives three regimes for the *fog index* $n \equiv -\mathrm{d}\ln\sigma_{\rm disc}/\mathrm{d}\ln X$:

- **background-free** ($\sum b \ll 1$): a few signal events suffice, $\sigma \propto X^{-1}$, $n = 1$;
- **Poisson-limited**: if $\boldsymbol{\varepsilon}_\perp \neq 0$, the first term dominates and $\sigma \propto X^{-1/2}$, $n = \frac{1}{2}$;
- **fog**: if the signal is (nearly) degenerate with a background ($\boldsymbol{\varepsilon}_\perp \approx 0$), q_0 saturates at a value set by the flux uncertainties r_k, and $\sigma_{\rm disc}$ stops improving: $n < \frac{1}{2}$, tending to 0.

O'Hare's index is the inverse of ours, $n_{\mathrm{O'H}} = 1/n$, so his fog condition $n_{\mathrm{O'H}} > 2$ corresponds to $n < \frac{1}{2}$ here. The question "can DM be seen through the neutrino background?" becomes "is the DM spectrum, after passing through $S(\vec{q},\omega)$, outside the span of the neutrino spectra?".

8 Directional signal

8.1 Rate for an anisotropic response

When S depends on $\hat{q}$, Eq. (2.4) with Eq. (2.5) gives

$$\frac{\mathrm{d}R}{\mathrm{d}\omega} = \frac{1}{\rho_T}\frac{\rho_\chi}{m_\chi}\frac{\sigma_n}{4\pi\mu^2}\int q\,\mathrm{d}q \int \frac{\mathrm{d}\Omega_q}{4\pi}\, 2g(v^*,\hat{q})\, S(\vec{q},\omega), \tag{8.1}$$

which reduces to Eq. (2.7) because $\int \frac{\mathrm{d}\Omega_q}{4\pi} 2g = \eta$ (Eq. (2.6)). For the boosted, truncated Maxwellian the Radon transform is analytic:

$$g(w,\hat{q}) = \frac{1}{N_{\mathrm{esc}}\sqrt{\pi}v_0}\Big[e^{-(w+\hat{q}\cdot\vec{v}_E)^2/v_0^2} - e^{-z^2}\Big]\Theta\big(v_{\mathrm{esc}} - |w+\hat{q}\cdot\vec{v}_E|\big). \tag{8.2}$$

It follows by integrating the Gaussian over the two velocity components perpendicular to $\hat{q}$ within the escape sphere. The rate depends on the angle θ_E between $\vec{v}_E$ and the crystal c axis. Since $D^{\hat{q}}$ is quadratic in $\hat{q}$, $S(\vec{q}) = S(-\vec{q})$ and $R(\theta_E) = R(\pi - \theta_E)$.

Numerics. S is tabulated at 12 independent Gauss–Legendre nodes in $\cos\theta_q$, averaged over 4 azimuths; the residual azimuthal dependence is at most 2–6% in $\langle 1/\omega\rangle$. The angular integral of g uses product quadrature: S is held constant within each node's cell, while g, which has a kink at v_{esc}, is integrated finely. With this scheme, $\langle 2g\rangle_\Omega = \eta$ to 3×10^{-4}.

8.2 Laboratory geometry and modulation

For an axis with equatorial coordinates (α_c, δ_c),

$$\cos\theta_E = \sin\delta_c\sin\delta_E + \cos\delta_c\cos\delta_E\cos(\alpha_c - \alpha_E), \qquad (\alpha_E,\delta_E) \simeq (21^{\mathrm{h}}12^{\mathrm{m}}, +48^\circ). \tag{8.3}$$

A vertical c axis at LNGS has $\delta_c = 42.45^\circ$ and α_c equal to the local sidereal time, so it sweeps $\theta_E \in [6^\circ, 90^\circ]$ every sidereal day.

For a modulation $R(t) = \bar{R}[1 + A\cos(\Omega t)]$ with known phase, the Fisher information on A per event is $\langle\cos^2\rangle = \frac{1}{2}$. Hence $\sigma_A = \sqrt{2/N}$, and a 3σ detection requires

$$N \simeq \frac{18}{A^2} \quad (\approx 10^4 \text{ events for } A = 4\%). \tag{8.4}$$

9 Implementation, fixes and validation

9.1 From v1 to v2

v1 problem	Fix and effect
CEvNS prefactor $G_F^2/4\pi$ with $Q_W/2$	$G_F^2/8\pi^2$ with $Q_W = N - (1-4s_W^2)Z$. v1 rates were $\pi/2$ too large (^{8}B benchmark 1.585 → 0.9998).
SHM η low by $\sqrt{\pi}/2$	Standard expression (DM benchmark 0.897 → 0.9988).
Threshold not applied	Bins below E_{th} removed after smearing.
W and T_n inconsistent; $n_{\max} = 15$	IR regularization and one T_1 for everything; $n_{\max} = 60$; T_n to 3 eV. The f-sum rule now holds to 1% (v1: 27–72% violation at $q = 3$–230 keV).
ρ_T inconsistent with V	$\rho_T = M_{\mathrm{cell}}/V$ (errors were $CaWO_4$ +1%, Al_2O_3 +9%, TeO_2 +8%).
Duplicated mass grid; stale cache	One log grid; cache keyed by a hash of the tensor.
Run time (minutes)	Vectorized S, FFT convolutions, exact Newton profile: seconds per step.

9.2 Phonon inputs

Material	Tensor and checks
$CaWO_4$	Rebuilt from phonopy force constants (v5, $32 \times 32 \times 16$ mesh). The previous file broke the crystal symmetry ($\langle\omega\rangle_W$ 28–38 vs 21.8 meV). The $2 \times 2 \times 1$ supercell gives one in-plane imaginary branch and acoustic slopes 20–40% low; 87% of W's $\langle 1/\omega\rangle$ comes from 2–10 meV. No Born charges.
Al_2O_3	Matches a rebuild with LO–TO correction to 0.1% in $\langle\omega\rangle$ and $\langle 1/\omega\rangle$. Sound speeds within about 5% (acoustic DOS about 20% high from DFT softening).
TeO_2	Atom order corrected (file stored O×8, Te×4 under Te-first labels). No force constants to validate.
$LiNbO_3$	Good along c; small in-plane imaginary pocket.
Si	Phonon database, 64-atom supercell: TA 4.86 vs 5.84 km/s (about 15% soft). Supersedes the 16-atom file (TA 40% soft).

9.3 Validation

- **Free-nucleus limit** (above 1 eV, all five materials): within 0.1%.
- **Sum rules** (Eqs. 3.12–3.13): within about 1% at all q.
- **Likelihood**: the Newton profile agrees with an independent L-BFGS-B minimization to $|\Delta q_0| < 2 \times 10^{-3}$.
- **Directional code**: the $\vec{v}_E$-averaged rate equals the isotropic rate to 10^{-4} (Si) and 0.3% ($CaWO_4$, a physical second-order effect of the anisotropy). Si shows zero modulation.
- **External benchmarks**: the Campbell-Deem *et al.* rates are reproduced with DarkELF; see the comparisons with Stratman & Lin (Sec. 10.3) and with Bento *et al.* (Sec. 10.4).

10 Results

All results use an exposure of 1 t·yr and $\hat{q} \parallel c$ unless stated otherwise.

10.1 Regimes in $CaWO_4$

Including the full multiphonon expansion instead of single-phonon emission ($n = 1$) improves the reach by up to 38× at $E_{\rm th} = 30$ meV (at 49 MeV). The improvement comes from emission involving W. The $n = 1$ "bump" near 70 MeV is a collapse of the single-phonon rate, not a spectral degeneracy.

Energy resolution $\sigma_E = E_{\rm th}/5$ has different effects at different thresholds:

- **30 meV**: the reach improves by about 6%, because more events fluctuate above threshold.
- **0.1 eV**: the reach worsens by up to 1.4× below 5 MeV. The threshold coincides with the top optical peak at about 110 meV.
- **1 eV**: below about 35 MeV the curve is set by far Gaussian tails and should not be quoted.

Correcting the $CaWO_4$ tensor weakened the low-mass reach by 1.5–1.65×.

10.2 Five materials

Below 5 MeV at 30 meV, Al_2O_3 is best. Relative to it, Si is weaker by 1.3×, $LiNbO_3$ by 3.6×, TeO_2 by 8× and $CaWO_4$ by 16×. Above 100 MeV, TeO_2 and $CaWO_4$ lead by 1.3–1.8×, thanks to the coherent A^2 (Fig. 1). The multiphonon gap tracks the heaviest, softest atom: $CaWO_4$ 38, TeO_2 16, $LiNbO_3$ 8, Si 5.6, Al_2O_3 2.9.

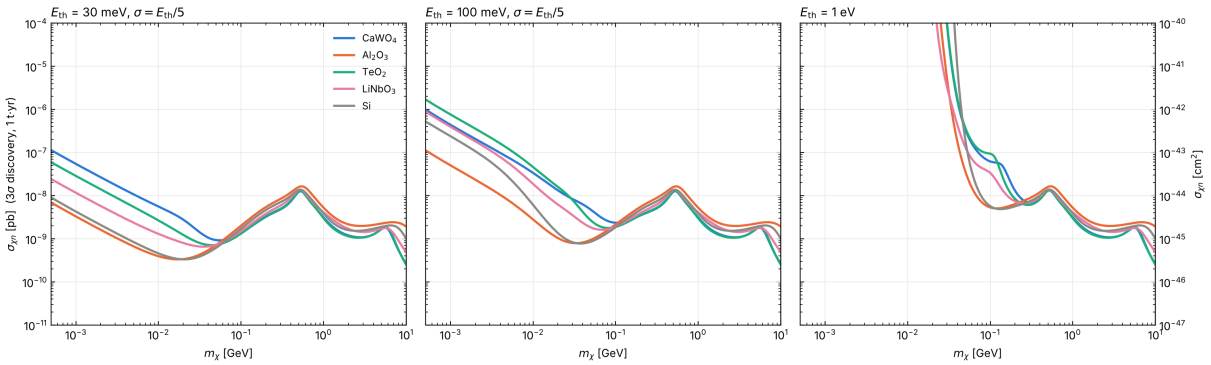


Figure 1: 3σ discovery limits (1 t·yr) for five crystals at $E_{\rm th} = 30\,\text{meV}$ and 100 meV (with $\sigma_E = E_{\rm th}/5$) and at 1 eV.

10.3 Daily modulation

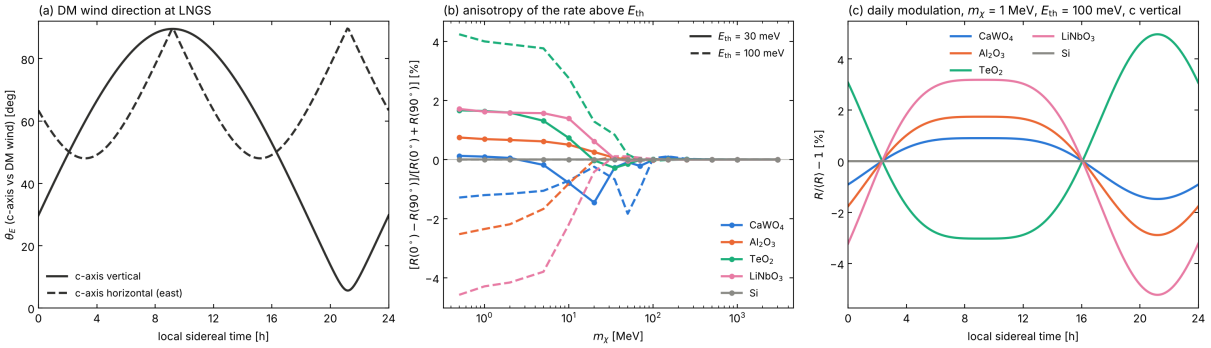


Figure 2: (a) θ_E over a sidereal day at LNGS. (b) Anisotropy $[R(0°) - R(90°)]/[R(0°) + R(90°)]$ vs mass at 30 meV (solid) and 100 meV (dashed). (c) Daily modulation at 1 MeV, 100 meV, for a vertical c axis.

The signed anisotropy at 1 MeV (30 / 100 meV) is:

- $CaWO_4$: +0.1% / −1.2%;
- Al_2O_3: +0.7% / −2.4%;
- TeO_2: +1.6% / +4.0%;
- $LiNbO_3$: +1.6% / −4.3%;
- Si: 0.

The effect disappears above about 50 MeV, where the impulse regime is isotropic. The sign reversal with threshold reflects the Poisson structure of Eq. (3.11):

- near threshold the rate weights $\hat{T}_1 \propto D/\omega$, favouring the softer direction;
- above the top optical phonon, events need several phonons from the top of the spectrum, favouring the stiffer direction.

Benchmark against Stratman & Lin. With their conventions ($v_E = 240$, $v_{\rm esc} = 500$ km/s, no resolution, $c \parallel \vec{v}_E$ at $t = 0$, $\theta_e = 42°$, $A_{\rm mod} = |R(0) - \langle R\rangle|/\langle R\rangle$), Al_2O_3 at 10 MeV gives $A_{\rm mod} = 2.7$, 5.3 and 7.4% for $E_{\rm th} = 120$, 200 and 300 meV, with a maximum of 9.8% at 3–5 MeV. Their result is up to about 11%. Magnitude, threshold dependence and mass range agree; the remaining 30% at 10 MeV is consistent with differences in the inputs.

10.4 Comparison with CRESST

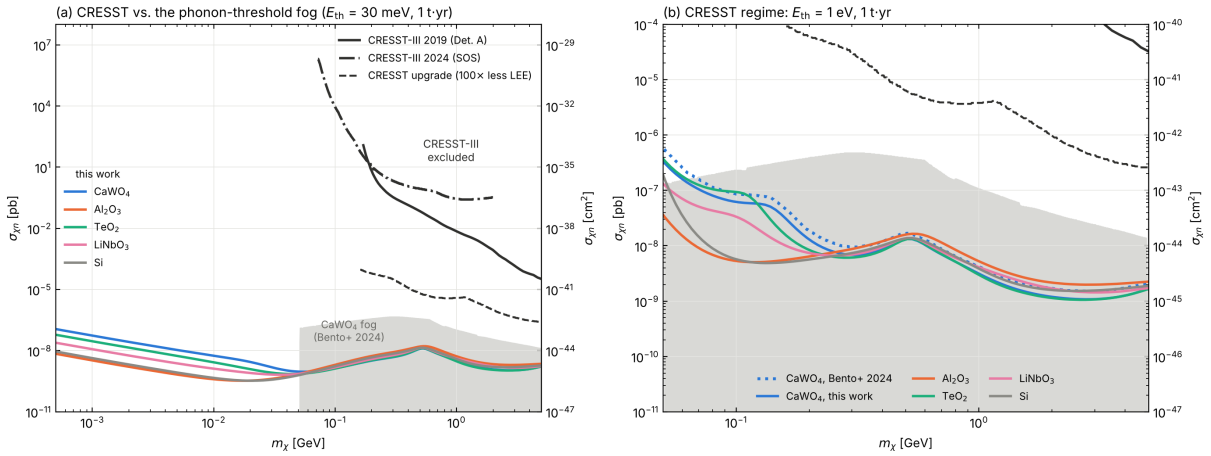


Figure 3: (a) CRESST-III limits (2019, published data; 2024 SOS, digitized) and the upgrade projection with 100× less low-energy excess, against the discovery limits of this work at 30 meV. Grey: $CaWO_4$ fog of Bento *et al.* (2024). (b) $E_{\rm th} = 1$ eV: this work vs Bento *et al.* for $CaWO_4$ (dotted).

- At 1 GeV, CRESST-III lies about six orders of magnitude above the neutrino-limited region, and the upgrade about two. CRESST is limited by its low-energy excess.
- A 30 meV phonon threshold opens the region from 0.5 MeV upward, which CRESST (lowest mass 73 MeV) does not reach. In that region Al_2O_3 and Si outperform $CaWO_4$ by more than an order of magnitude.
- At 1 eV our $CaWO_4$ curve reproduces the shape of Bento *et al.*, including the ^{8}B feature at 0.5 GeV, and lies a factor 0.61–0.78 below it.

10.5 Background or fog?

Above 30 meV, Al_2O_3 sees 3.7×10^3 neutrino events per t·yr (pp 52%, ^{7}Be 37%, ^{8}B 3%); $CaWO_4$ sees 2.8×10^4. Figure 4 shows:

- **1–20 MeV**: $n \to 0.50$ and never lower. The multiphonon DM spectrum has a component outside the span of the pp and ^{7}Be CEvNS spectra ($\varepsilon_\perp \neq 0$, Sec. 7.2). DM remains discoverable, and the reach improves as $X^{-1/2}$.
- **About 100 MeV**: mild saturation ($n \simeq 0.43$–0.47) above a few t·yr.
- **About 0.5 GeV**: a genuine fog, $n \simeq 0.2$–0.4, where the DM spectrum is nearly degenerate with ^{8}B.

Representative values for Al_2O_3: $\sigma_{\rm disc}(1\,{\rm MeV}) = 2.8 \times 10^{-43}\,{\rm cm}^2$ at 1 kg·yr and $3.3 \times 10^{-45}\,{\rm cm}^2$ at 1 t·yr.

To reach the neutrino-limited regime, a detector needs:

- a threshold of about 30 meV with $\sigma_E \approx 6$ meV;
- an exposure of about 0.1–1 t·yr (Al_2O_3);
- non-neutrino backgrounds below about 0.01 events/(kg·day) above threshold.

The last requirement is many orders of magnitude below today's low-energy excess.

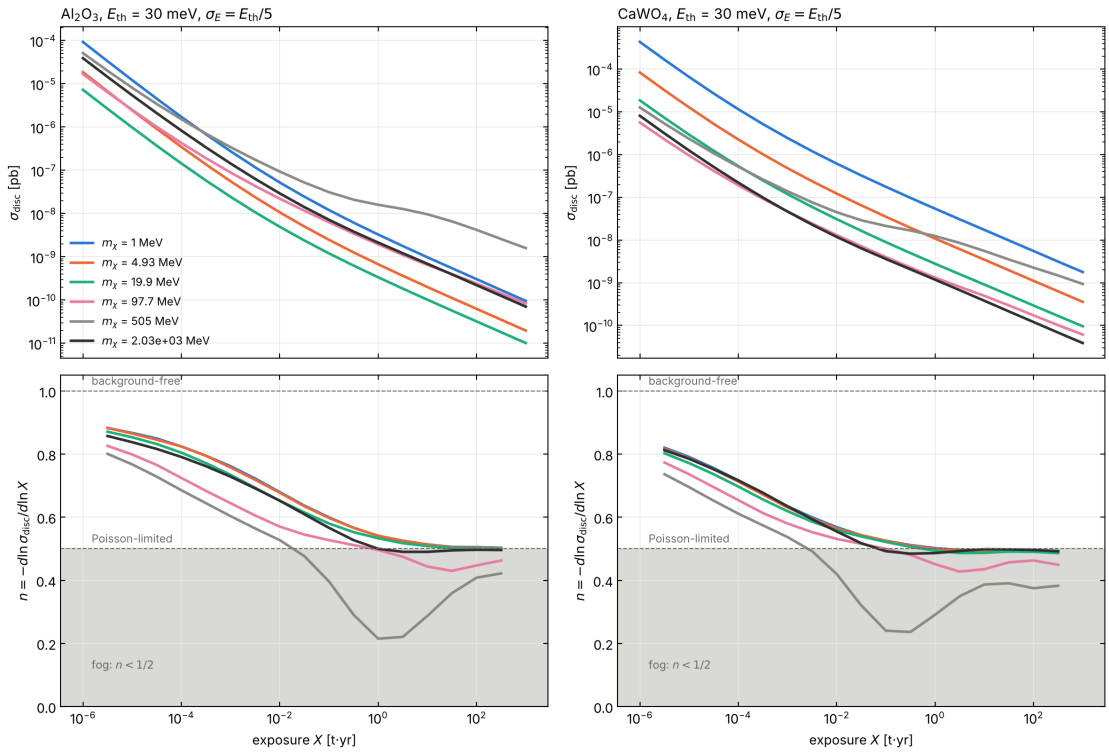


Figure 4: Discovery cross section (top) and fog index $n = -\mathrm{d}\ln\sigma/\mathrm{d}\ln X$ (bottom) vs exposure, at $E_{\rm th} = 30\,\mathrm{meV}$ with $\sigma_E = E_{\rm th}/5$. The grey band is the fog, $n < \frac{1}{2}$.

11 Why the lattice response must be known precisely

11.1 Power-law sensitivity of the multiphonon rate

By Eq. (3.11), the n-phonon contribution is $\mathrm{Pois}(n;\lambda)\hat{T}_n(\omega)$ with $\lambda = q^2\langle 1/\omega\rangle/2m_d$. Holding the shape $\hat{T}_1$ fixed and varying the moment $\langle 1/\omega\rangle$,

$$\frac{\partial \ln S_{d,n}}{\partial \ln\langle 1/\omega\rangle} = n - \lambda. \tag{11.1}$$

At low DM mass q is small and $\lambda \ll 1$, so the n-phonon rate scales as $\langle 1/\omega\rangle^n$. A deposit above threshold needs at least

$$n_{\rm min} = \left\lceil \frac{E_{\rm th}}{\omega_{\rm max}} \right\rceil \tag{11.2}$$

phonons, where $\omega_{\rm max}$ is the top of the phonon spectrum. The rate above threshold is therefore $\propto \langle 1/\omega\rangle^{n_{\rm min}}$. For example, a 20% error in $\langle 1/\omega\rangle$ (the size of the acoustic-slope errors found in the $CaWO_4$ and Si force constants) changes a three-phonon rate by $1.2^3 \simeq 1.7$. That is exactly the size of the change observed when the $CaWO_4$ tensor was corrected (1.5–1.65×).

Because $\langle 1/\omega\rangle$ is dominated by the softest modes (87% of the tungsten value comes from 2–10 meV), the low-energy acoustic branches control the low-mass reach. Those are the branches most affected by supercell size, imaginary modes and DFT softening.

11.2 Sensitivity to the top of the spectrum

The support of $\hat{T}_n$ ends at $n\,\omega_{\rm max}$. When $E_{\rm th}$ is close to a multiple of $\omega_{\rm max}$, a small shift of the highest optical branch changes $n_{\rm min}$ by one, and the rate by a factor of order λ, which can be orders of magnitude at low mass. The shape of $D(\omega)$ near the band top sets how the rate falls across threshold. This is why the $CaWO_4$ reach at $E_{\rm th} = 0.1\,\mathrm{eV}$, right at the 110 meV optical peak, changes by 1.4× when the edge is smeared by 20 meV resolution.

11.3 Discrimination depends on the spectral shape

Section 7.2 shows that DM can be separated from neutrinos only through the component of its spectrum orthogonal to the neutrino templates. Both spectra are produced by the same $S(\vec{q}, \omega)$, with different weights in q:

- DM: $q \lesssim 2m_\chi v$, weighted by $\eta(v^*)$;
- CEvNS: $q \lesssim 2E_\nu$, weighted by the neutrino spectrum.

The orthogonal component, and with it the $n = \frac{1}{2}$ behaviour in Fig. 4, is a property of the lattice response. If S is uncertain, its uncertainty enters the likelihood as a further nuisance direction that moves signal and background together. Any part of the DM spectrum that this nuisance can mimic becomes degenerate. That converts a Poisson background into a fog, $n < \frac{1}{2}$, even where the neutrino fluxes alone would not. An imprecise S therefore does more than shift the curves: it can change the scaling of the reach with exposure.

11.4 Further consequences

- **Target selection.** The 16× spread between materials at low mass and the 2.9–38× multiphonon gap are properties of $D_d(\omega)$.
- **Directionality** requires the per-atom anisotropy of the DOS tensor, and therefore the eigenvectors, not only the frequencies. Its sign flips with threshold.
- **Neutrino background.** pp and ^{7}Be CEvNS populate the same meV–eV window through the same S, so the background prediction inherits the same uncertainties.

11.5 Sources of uncertainty and how to control them

- **Force constants**: supercell convergence (soft TA branches, imaginary pockets), DFT functional (acoustic softening), missing LO–TO correction. *Control:* converged supercells, Born charges, comparison with measured sound speeds and inelastic neutron scattering DOS.
- **Model approximations**: harmonic and $T = 0$ approximations; incoherent approximation (invalid for $q \lesssim$ few keV); impulse switch at $\lambda = 16$. *Control:* coherent single-phonon calculations at low q; anharmonic corrections where n is large.
- **Propagation**: include a parametrized uncertainty on S (for example, the scale of $\langle 1/\omega \rangle$ and of the acoustic slopes) as a nuisance in the likelihood. This quantifies how much of the low-mass reach is robust.

12 Conclusions

1. The rewritten pipeline reproduces the free-nucleus limit to 0.1% and the phonon sum rules to 1%, for five materials. It runs in seconds.
2. Multiphonon emission dominates the low-mass reach; ignoring it costs up to 38× in $CaWO_4$. For thresholds of tens of meV, Al_2O_3 and Si are the best targets below 50 MeV, while heavy targets win above 100 MeV.
3. In the phonon regime (1–20 MeV) the neutrino background is a Poisson background, not a fog. DM is discoverable through it, and the reach improves as $X^{-1/2}$. A genuine fog appears only near 0.5 GeV (^{8}B), with mild saturation from about 100 MeV.
4. CRESST is 2–6 orders of magnitude above this regime and limited by its low-energy excess. Reaching the neutrino-limited phonon regime requires thresholds of about 30 meV and backgrounds below about 0.01/(kg·day).

5. Anisotropic crystals show a percent-level sidereal modulation (up to about 10% at higher thresholds) below about 50 MeV, with a threshold-dependent phase. It is a tool to confirm the DM origin of a signal, not to discover one.

6. All these conclusions depend on $S(\vec{q}, \omega)$. The multiphonon rate scales as a power of the phonon moments, and the separation of DM from neutrinos depends on the spectral shape. Converged, experimentally validated phonon inputs, and their uncertainty propagated into the likelihood, are therefore required for robust low-mass projections. The most urgent case is $CaWO_4$.

References

[1] CRESST Collaboration, Commun. Phys. (2025), doi:10.1038/s42005-025-02476-5.

[2] A. Bento *et al.*, Eur. Phys. J. C **84**, 1118 (2024).

[3] B. Campbell-Deem, S. Knapen, T. Lin, E. Villarama, Phys. Rev. D **106**, 036019 (2022).

[4] S. Knapen, J. Kozaczuk, T. Lin, Phys. Rev. D **105**, 015014 (2022) [DarkELF].

[5] C. Stratman, T. Lin, arXiv:2411.03433.

[6] S. Griffin, S. Knapen, T. Lin, K. M. Zurek, Phys. Rev. D **98**, 115034 (2018).

[7] C. A. J. O'Hare, Phys. Rev. Lett. **127**, 251802 (2021).

[8] A. H. Abdelhameed *et al.* (CRESST), Phys. Rev. D **100**, 102002 (2019).

A Program inventory (NuFog_v2)

Program	Purpose
`config.py`	Settings (material via `NUFOG_MATERIAL`, grids, $n_{\max}$, $E_{\rm th}$, exposure, resolution, halo)
`nufog_core.py`	$S(\vec{q}, \omega)$, DM and CEvNS rates, binning, resolution
`likelihood.py`	Asimov q_0 with Newton profile
`step1`–`step4`, `run_all.py`	Neutrino spectra, DM matrix, discovery curves, plot
`check_normalization.py`, `compare_v1.py`	Free-nucleus benchmark; v1 vs v2
`diagnose_spectra.py`, `compare_materials.py`	Spectra diagnostics; material comparison
`directional.py`, `check_directional.py`	$R(m_\chi, \theta_E)$; isotropic-limit check
`daily_modulation.py`, `directional_summary.py`, `plot_directional.py`	Sidereal modulation and figure
`compare_stratman_lin.py`	Benchmark against Stratman & Lin
`plot_fog_cresst.py`, `literature/`	CRESST comparison (published and digitized curves)
`exposure_scan.py`, `plot_exposure_scan.py`	Fog index vs exposure